

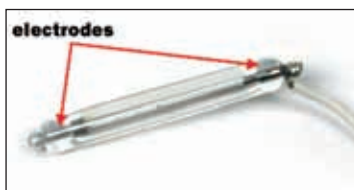
tube which looks like a neon light to produce the actual flash and an electronic circuit connecting them. The tube has a metal trigger in the middle and is filled with xenon gas.

The idea is simple enough; electrons in the electrical current move through the gas and energise the gas atoms, causing the gas to glow

white. Modelling lights are common items around studios. They are basically an incandescent bulb close to the flash tube providing continuous illumination to subjects. For specialised purposes like capturing extremely fast moving objects or reactions, micro-flashes are available which are high voltage flash units discharging a flash of quick light with an exceptionally quick, sub-microsecond duration.

Though it seems simple, flashes prove to be one of the most widely misused camera equipments in photography. It is important to remember that when you use a flash as compared to normal camera exposure, you are adding basically handling two light sources- the ambient light, in which you can visualize the capture before clicking, and your flash, in which you can control and vary the intensity, position and distribution of light but not see the result before the actual shot. Thus you are in a way controlling factors that otherwise would have required changing the exposure or depth of field, but you have to require extra intuition to achieve a desired affect and look without checking your camera. Thus it is critical to have a good understanding of how flashes change the appearance of a subject qualitatively.

Lets look at a few accompanying terms you will most likely come across when dealing with flashes. Say for instance, Flash Ratio, which describes the mix between light from the flash and the surroundings. Flash ratios are controlled by modifying shutter speeds as they affect the ambient light entering the camera but not the light from the flash. Also, for a given amount of ambient light, this value is controlled by playing with two variables, flash intensity and length of exposure. One of the toughest tasks when working with flashes is understanding how different camera flash metering modes will affect the overall exposure. A simplified guide will be to treat the auto mode as having a flash ratio of 1:1 or greater if dim, else no flash will be fired from the camera. Program mode is similar to auto, except that it allows you to use the flash at your will, letting you forcefully use it in



Typical camera flash tube, removed from the housing